Information for Candidates

Numerical Answers

- Numerical answers should be given as fully evaluated decimal values and not as unevaluated arithmetic expressions.

Notation

The following notation is used in this paper:

- The voltage waveform at node X in an AC circuit is denoted by $x(t)$, while the phasor voltage is denoted by X.
- Component and source values are generally given in Ohms, Farads, Henrys, Volts, or Amps, unless otherwise stated.
- Times are in seconds, unless otherwise stated.

1. (a) For the circuit shown in Figure 1a, use the principle of superposition to determine the Thevenin equivalent circuit as seen from nodes X and Y. [6]

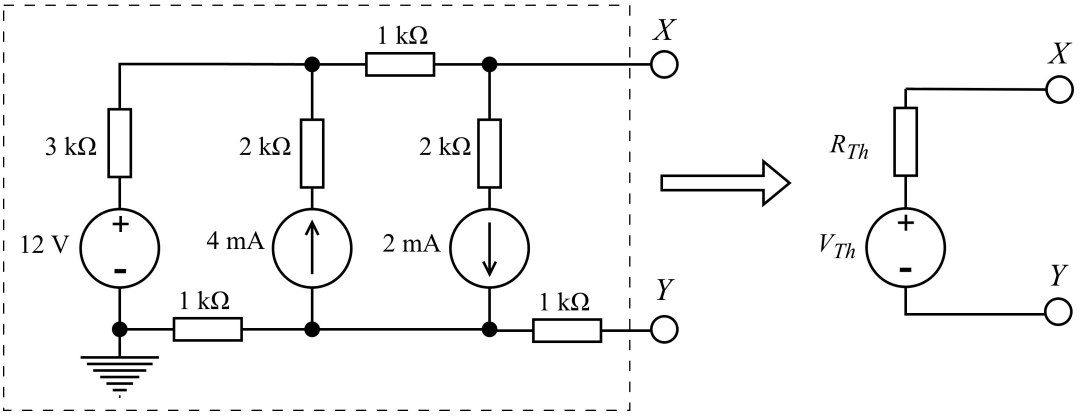


Figure 1a

Solution:

(2 marks) Find R_{Th}

All independent sources are turned off: the voltage source is shorted and the current sources are opened.

The resistance seen between X and Y is

$$R_{\text{Th}} = 1\,\text{k}\Omega + 3\,\text{k}\Omega + 1\,\text{k}\Omega + 1\,\text{k}\Omega = 6\,\text{k}\Omega.$$

(4 marks) Find V_{Th} using superposition

$$V_{\text{Th}} = V_1 + V_2 + V_3$$

Case 1: Only the 12 V source active

$$V_1 = 12\,\text{V}$$

Case 2: Only the 4 mA source active

$$V_2 = \frac{4}{6}\,(4\,\text{mA} \times 6\,\text{k}\Omega) = 16\,\text{V}$$

Case 3: Only the 2 mA source active

$$V_3 = -\frac{5}{7}\,(2\,\text{mA} \times 7\,\text{k}\Omega) = -10\,\text{V}$$

$$V_{\text{Th}} = 12 + 16 - 10 = 18\,\text{V}$$

$$\boxed{V_{\text{Th}} = 18\,\text{V}}, \qquad \boxed{R_{\text{Th}} = 6\,\text{k}\Omega}$$

(b) For the circuit shown in Figure 1b, the input voltage is [4+3]

$$V_{in}(t) = 400\cos(100\pi t)\ \text{V}.$$

i. With the capacitor branch open-circuited (C_P disconnected), determine the output voltage $V_{out}(t)$, the load active power P_L, and the load reactive power Q_L.

ii. The capacitor C_P is then connected in parallel with the load. Determine the new output voltage $V_{out}(t)$. Only the steady-state voltage is required. The transient can be ignored.

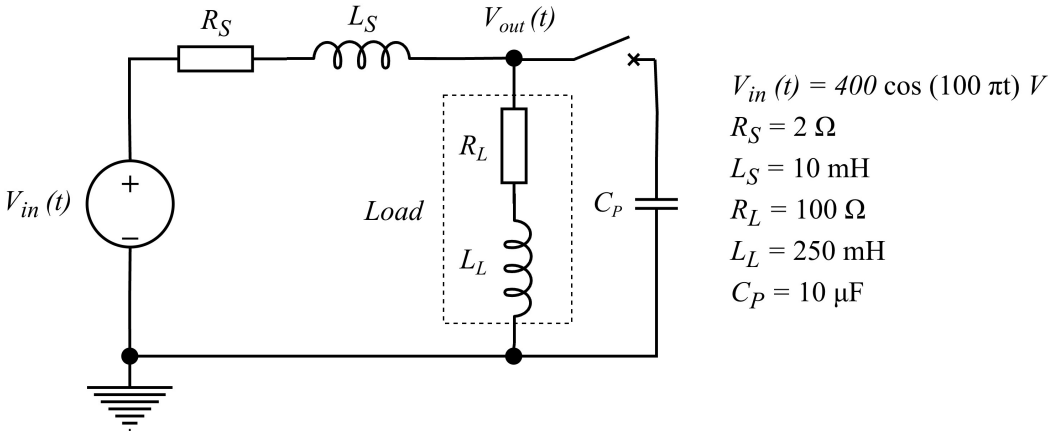


Figure 1b

Solution:

(4 marks) With C_P disconnected

At

$$\omega = 100\pi = 314.16 \text{ rad s}^{-1},$$

the source and load impedances are

$$Z_S = R_S + j\omega L_S = 2 + j3.14\ \Omega, \qquad Z_L = R_L + j\omega L_L = 100 + j78.54\ \Omega.$$

Hence

$$Z_{\text{tot}} = Z_S + Z_L = 102 + j81.68\ \Omega.$$

The circuit current is

$$I = \frac{400\angle 0^\circ}{102 + j81.68} = 3.06\angle -38.7^\circ \text{ A}.$$

Therefore

$$\boxed{V_{out} = IZ_L = 389.2\angle -0.54^\circ \text{ V}},$$

so

$$V_{out}(t) = 389.2\cos(100\pi t - 0.54^\circ) \text{ V}.$$

The RMS current is

$$I_{\text{rms}} = \frac{3.06}{\sqrt{2}} = 2.16 \text{ A}.$$

The load complex power is

$$S_L = |I_{\text{rms}}|^2 Z_L = (2.16)^2(100 + j78.54) = 468.5 + j368.0 \text{ VA}.$$

Hence

$$\boxed{P_L = 468.5 \text{ W}}, \qquad \boxed{Q_L = 368.0 \text{ var}}.$$

(3 marks) With C_P connected
The capacitor impedance is

$$Z_C = \frac{1}{j\omega C_P} = -j318.3 \ \Omega.$$

The new load impedance is

$$Z'_L = Z_L \parallel Z_C = \frac{(100 + j78.54)(-j318.3)}{100 + j78.54 - j318.3} = 150.1 + j41.7 \ \Omega.$$

Hence

$$V'_{out} = 400\angle 0° \frac{150.1 + j41.7}{(2 + j3.14) + (150.1 + j41.7)} = 393.0\angle -0.90° \text{ V}.$$

Therefore

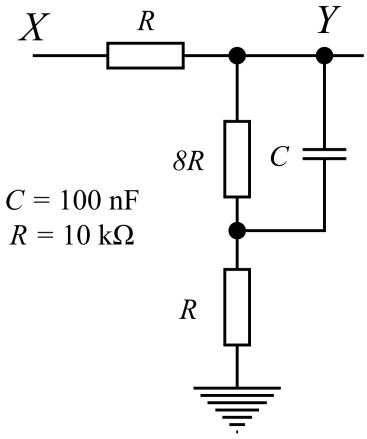
$$\boxed{V_{out}(t) = 393.0 \cos(100\pi t - 0.90°) \text{ V}}.$$

(c) Figure 1c shows the circuit and the input signal used in this question. [2+5]

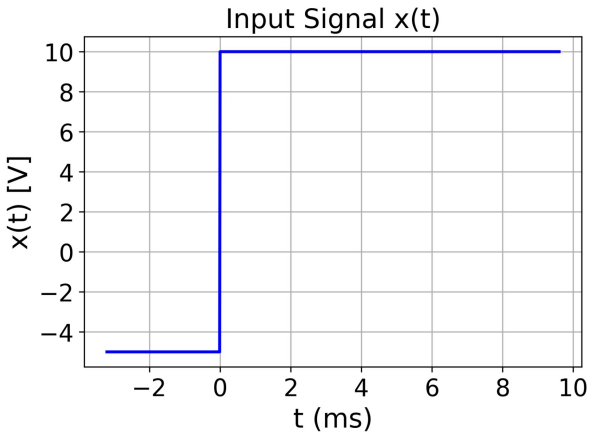
i. For the circuit shown in Figure 1c - (i), determine the transfer function:

$$H(j\omega) = \frac{Y}{X}$$

ii. The input voltage $x(t)$ changes from -5 V to $+10$ V at $t = 0$, as illustrated in Figure 1c - (ii). Assuming the circuit is in steady state for $t < 0$, determine the output voltage $y(t)$ for $t \geq 0$.



(i)



(ii)

Figure 1c

Solution:

(2 marks) $H(j\omega)$:

$$Z_C = \frac{1}{j\omega C}$$

$$Z_p = 8R \parallel Z_C = \frac{8R\, Z_C}{8R + Z_C} = \frac{8R}{1 + j\omega 8RC}$$

$$H(j\omega) = \frac{Y}{X} = \frac{R + Z_p}{2R + Z_p}$$

$$H(j\omega) = \frac{R + \dfrac{8R}{1 + j\omega 8RC}}{2R + \dfrac{8R}{1 + j\omega 8RC}} = \frac{9 + j\omega 8RC}{10 + j\omega 16RC}$$

Therefore,

$$\boxed{H(j\omega) = \frac{9 + j\omega 8RC}{10 + j\omega 16RC}}$$

With $R = 10$ kΩ and $C = 100$ nF,

$$\boxed{H(j\omega) = \frac{9 + j0.008\,\omega}{10 + j0.016\,\omega}}$$

(5 marks) $y(t)$ for $t > 0$:

$$R_{\text{th}} = 8R \parallel (R + R) = 8R \parallel 2R = \frac{8R}{5}$$

$$\tau = R_{\text{th}} C = \frac{8R}{5} C$$

$$\tau = \frac{8}{5}(10^4)(100 \times 10^{-9}) = 1.6 \times 10^{-3} \text{ s} = 1.6 \text{ ms}$$

For $t < 0$, the capacitor is open-circuit, so the three resistors are in series: Since $x(0^-) = -5$ V, hence

$$y(0^-) = -5 + \frac{5}{10} = -4.5 \text{ V}$$

Since capacitor voltage cannot change instantaneously,

$$v_C(0^+) = v_C(0^-) = -4 \text{ V}$$

Also,

$$H(\infty) = \lim_{\omega \to \infty} H(j\omega) = \frac{8RC}{16RC} = \frac{1}{2}$$

The input changes by

$$\Delta x = 10 - (-5) = 15 \text{ V}$$

Hence the output changes instantaneously by

$$\Delta y = H(\infty)\Delta x = \frac{1}{2}(15) = 7.5 \text{ V}$$

Therefore,

$$y(0^+) = y(0^-) + \Delta y = -4.5 + 7.5 = 3 \text{ V}$$

Also,

$$H(0) = \frac{9}{10}$$

so the final value is

$$y(\infty) = H(0) \times 10 = 9 \text{ V}$$

Thus

$$y(t) = y(\infty) + \left[y(0^+) - y(\infty)\right]e^{-t/\tau}$$

$$y(t) = 9 + (3 - 9)e^{-t/1.6 \text{ ms}}$$

$$\boxed{y(t) = 9 - 6e^{-t/1.6 \text{ ms}} \text{ V}, \qquad t \geq 0}$$

(d) The amplifier in Figure 1d was originally designed to operate from a 12 V supply at room temperature with a gain of -240. Calculate the supply voltage required for it to produce a gain of -120, and the quiescent collector voltage under that new supply voltage. [7]

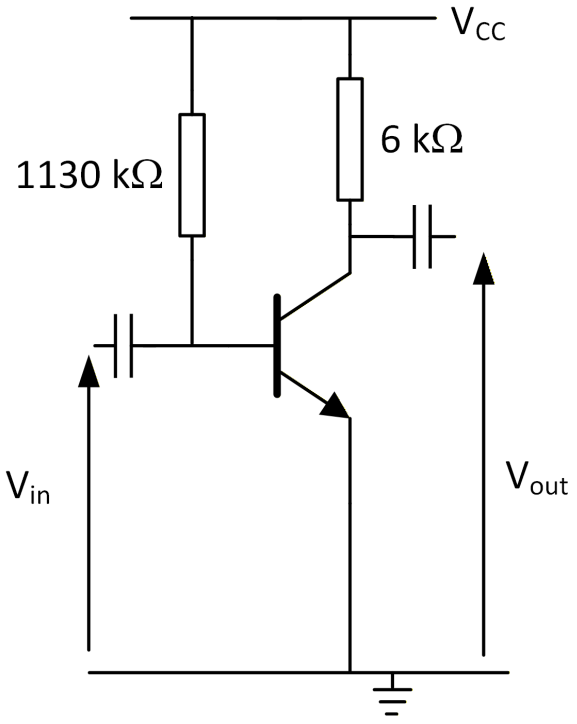


Figure 1d

With a 12 V supply, the base current is (12-0.7)/1130k=0.01 mA. This gives a collector current of 1 mA, and hence a value of gm at room temp of 40 mS. The gain is therefore -40m $\times$ 6k=-240, as stated in the question.

In order to achieve a gain of -120, we need to halve the value of gm and hence halve the value of collector current to 0.5 mA.

This requires a base current of 0.005 mA. The base current is given by $(V_{CC} - 0.7)/R_B$ and hence V_{CC}=6.35 V.

Under this condition the quiescent collector voltage is 6.35-0.5×3-3.35 V.

(e) Calculate the small signal gain of the circuit shown in Figure 1e. Assume R_G is large. [6]

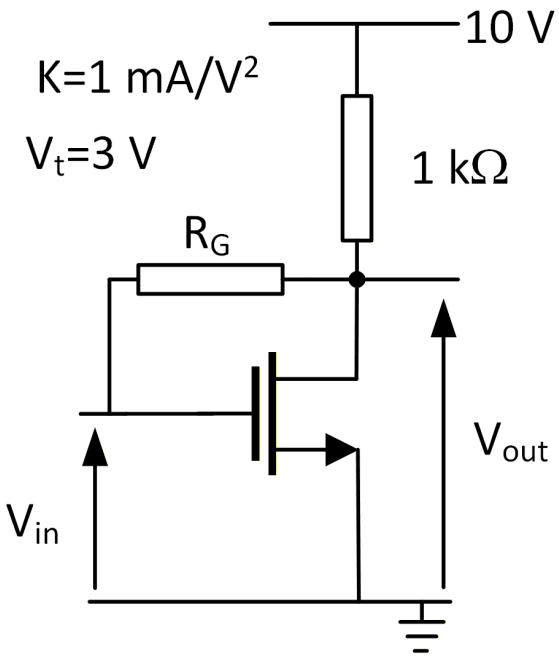


Figure 1e

We first need to calculate the quiescent operating point of the circuit. An enhancement mode n-channel mosfet with with gate biased from the drain is in active mode.

the drain current is:

$$I_D = K(V_{GS} - V_t)^2 = K(V_{out} - V_t)^2$$

We also know that:

$$I_D = (10 - V_{out})/R_D$$

Equating the two:

$$K(V_{out} - V_t)^2 = (10 - V_{out})/R_D$$

$$V_{GS} = V_{out} = 5.19 or -0.19.$$

Thus V_{GS}=5.19 V
And so $gm = 2K(V_{GS} - V_t) = 2 \times 1 \times (5.19 - 3) = 4.38mS$
This gives a gain of $-gmR_D$=-4.38

(f) Explain why the biasing scheme shown in the circuit on the left of Figure 1f is a poor practical solution, but the scheme shown in the circuit on the right (assuming the appropriate choice of component values) is a good solution. [7]

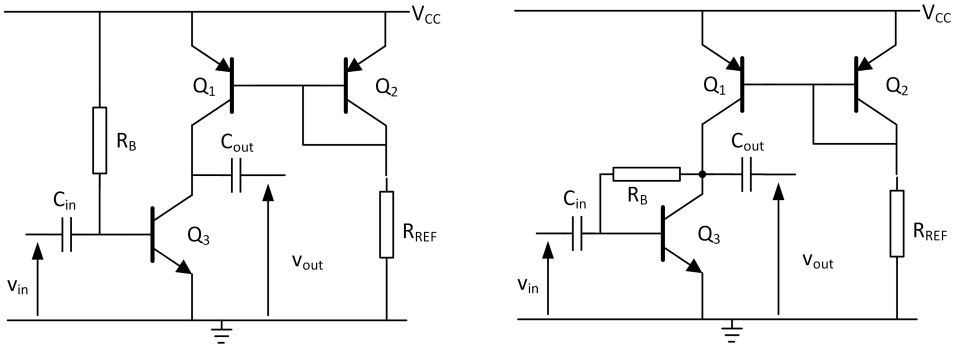


Figure 1f

Solution:

Both circuits show a common emitter amplifier configuration using an active load constructed using a current mirror. The left circuit tries to bias the amplifying transistor with a fixed base current (set by R_B), hence trying to force a set collector current of β times that. At the same time, the output of the simple current mirror is also trying to set a collector bias current condition that is controlled by R_{REF}. It is very difficult in practise to make these two currents identical and hence very difficult to keep both transistors in the forward active mode. You may think of this circuit as two current sources in series - and their currents can never be exactly matched.

The circuit on the right biases Q_2 using feedback from the collector. Hence in that circuit, the current mirror sets the bias condition and transistor Q_3 matches this condition automatically through the negative feedback set up by R_B.

2. The circuit shown in Figure 2 is an active filter designed for speech processing. Using this circuit, answer the following questions.

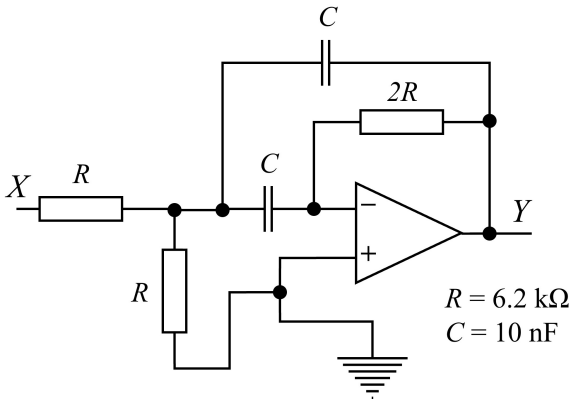


Figure 2

(a) Assuming that the operational amplifier is ideal, use the golden rules of the op-amp to show that the transfer function of the circuit is [8]

$$H(j\omega) = \frac{Y}{X} = -\frac{j\omega RC}{(j\omega RC)^2 + j\omega RC + 1}.$$

Solution:

(8 marks) $H(j\omega)$:

Let the node between the two resistors and the two capacitors be V_A.

Applying KCL at the inverting input node,

$$j\omega C(V_A - 0) + \frac{Y - 0}{2R} = 0$$

so,

$$j\omega C\, V_A + \frac{Y}{2R} = 0$$

Hence,

$$Y = -2Rj\omega C\, V_A$$

Applying KCL at node V_A,

$$\frac{V_A - X}{R} + \frac{V_A}{R} + j\omega C(V_A - 0) + j\omega C(V_A - Y) = 0$$

$$\frac{2V_A - X}{R} + j\omega C(2V_A - Y) = 0$$

Substituting

$$Y = -2Rj\omega C\, V_A$$

into the above equation,

$$\frac{2V_A - X}{R} + j\omega C\,(2V_A + 2Rj\omega C\, V_A) = 0$$

Multiplying throughout by R,

$$2V_A - X + 2j\omega RC\,V_A + 2(j\omega RC)^2 V_A = 0$$

$$X = 2V_A\left[1 + j\omega RC + (j\omega RC)^2\right]$$

Therefore,

$$H(j\omega) = \frac{Y}{X} = \frac{-2Rj\omega C\,V_A}{2V_A\left[1 + j\omega RC + (j\omega RC)^2\right]}$$

$$\boxed{H(j\omega) = \frac{Y}{X} = -\frac{j\omega RC}{(j\omega RC)^2 + j\omega RC + 1}}$$

(b) From the transfer function $H(j\omega)$ given in Part (a), find expressions for the low-frequency and high-frequency asymptotes. Also, determine the corner frequency in both radians per second (rad/s) and hertz (Hz). [6]

Solution:

Given

$$H(j\omega) = \frac{Y}{X} = -\frac{j\omega RC}{(j\omega RC)^2 + j\omega RC + 1}$$

Low-frequency asymptote (LF):

At low frequencies, take the lowest power of $j\omega$ in the numerator and denominator. Hence,

$$\boxed{H_{LF}(j\omega) \approx -j\omega RC}$$

High-frequency asymptote (HF):

At high frequencies, take the highest power of $j\omega$ in the numerator and denominator. Hence,

$$\boxed{H_{HF}(j\omega) \approx -\frac{1}{j\omega RC}}$$

Corner frequency:

The corner frequency occurs when

$$\omega RC = 1$$

Therefore,

$$\omega_c = \frac{1}{RC}$$

$$\boxed{\omega_c \approx 1.61 \times 10^4 \text{ rad/s}}$$

$$\boxed{f_c \approx 2.57 \times 10^3 \text{ Hz}}$$

(c) By comparing the denominator of $H(j\omega)$ with the standard second-order form, [4]

$$\left(\frac{j\omega}{\omega_0}\right)^2 + 2\zeta\left(\frac{j\omega}{\omega_0}\right) + 1,$$

determine the damping ratio ζ and hence the quality factor Q, given that $Q = \frac{1}{2\zeta}$.

Solution:

(4 marks) Given

$$H(j\omega) = -\frac{j\omega RC}{(j\omega RC)^2 + j\omega RC + 1}$$

The denominator can be written in the standard second-order form

$$\left(\frac{j\omega}{\omega_0}\right)^2 + 2\zeta\left(\frac{j\omega}{\omega_0}\right) + 1$$

Since

$$\omega_0 = \frac{1}{RC},$$

we have

$$j\omega RC = \frac{j\omega}{\omega_0}.$$

Thus the denominator becomes

$$\left(\frac{j\omega}{\omega_0}\right)^2 + \left(\frac{j\omega}{\omega_0}\right) + 1.$$

Comparing with the standard form,

$$2\zeta = 1$$

$$\boxed{\zeta = \frac{1}{2}}$$

Using

$$Q = \frac{1}{2\zeta},$$

$$\boxed{Q = 1}.$$

(d) Based on your answers to Parts (b) and (c), state whether the filter is critically damped, underdamped, or overdamped. Also, comment on whether the corner frequency is suitable for use in a speech-processing filter. [4]

Solution:

(4 marks) Since $0 < \zeta < 1$, the system is **underdamped**. An underdamped second-order system exhibits a resonant behaviour, which is consistent with the band-pass response of the filter.

The corner (centre) frequency lies within the main speech frequency band (roughly 300 Hz to 3 kHz), where most of the intelligibility of speech is concentrated. Therefore, the chosen corner frequency is suitable for speech processing because it allows the filter to emphasise important speech components while attenuating frequencies outside the speech band.

(e) Using the results from Part (b) and (c), sketch a straight-line (asymptotic) approximation of the frequency response between 100 Hz and 100 kHz. Only the magnitude plot is required. [8]

Solution:

(8 marks) Using the results from Parts (b) and (c), the circuit has a band-pass magnitude response.

At low frequencies,

$$|H(j\omega)| \propto \omega$$

so the straight-line magnitude plot rises with slope

$$+20 \text{ dB/decade.}$$

At high frequencies,

$$|H(j\omega)| \propto \frac{1}{\omega}$$

so the straight-line magnitude plot falls with slope

$$-20 \text{ dB/decade.}$$

The corner (resonant) frequency is

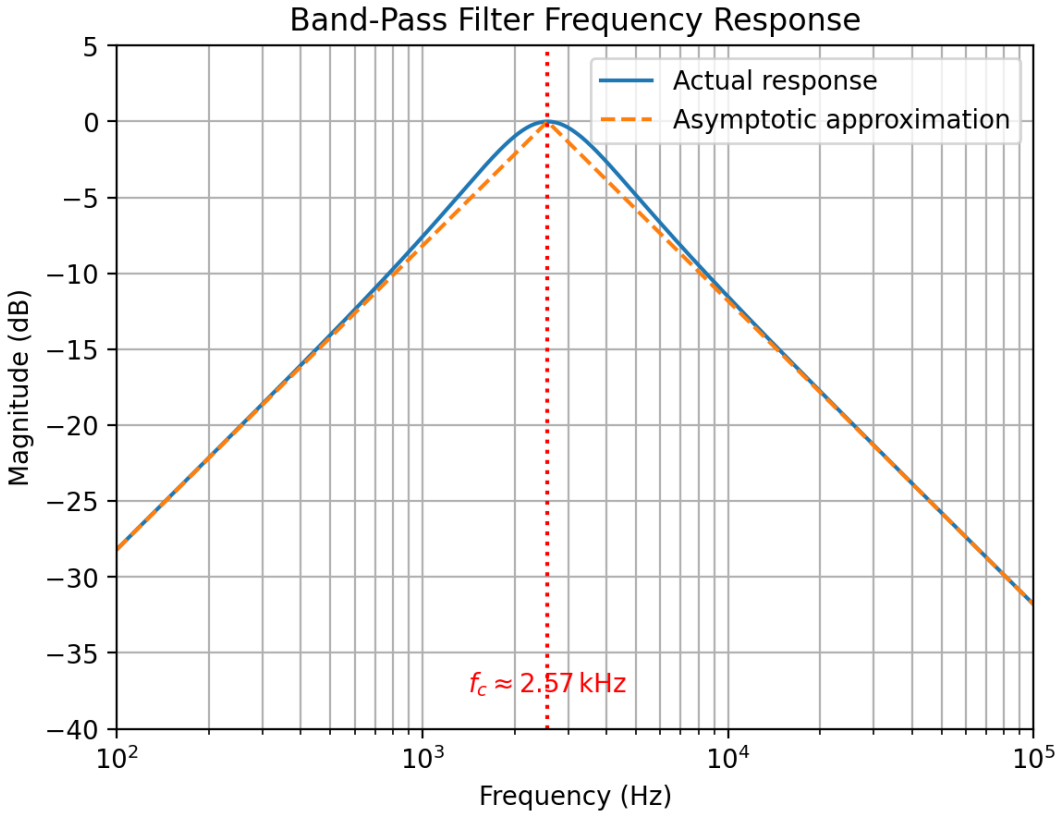
$$\omega_c \approx 1.61 \times 10^4 \text{ rad/s}$$

or

$$f_c \approx 2.57 \times 10^3 \text{ Hz.}$$

Hence, the asymptotic Bode magnitude plot:

- rises at +20 dB/decade from 100 Hz to f_c,
- has its break point at $f_c \approx 2.57$ kHz,
- falls at −20 dB/decade from f_c to 100 kHz.



Asymptotic Bode magnitude plot for Part 2e

3. A differential amplifier shown in Figure 3. Assume all MOSFETs are identical.

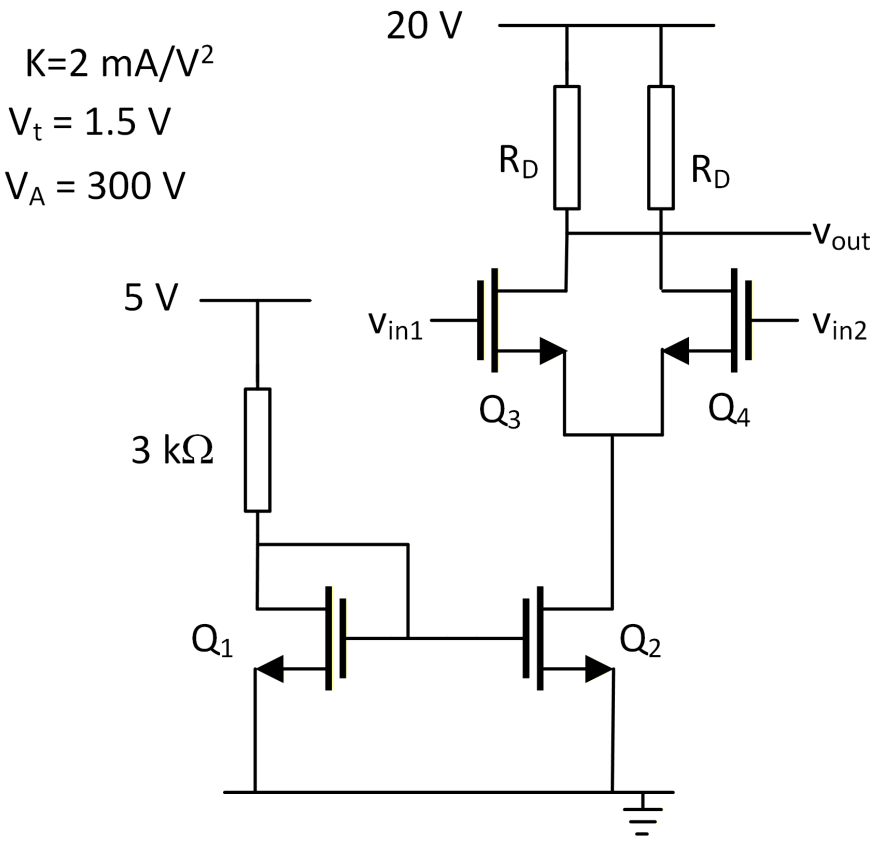


Figure 3

(a) Explain how the differential amplifier operates. In your answer be sure to describe how the circuit can amplify differential signals and (largely) reject common mode signals. [5]

Solution:

Q1 and Q2 act as a simple current mirror and hence Q2 pulls an (almost) constant current out of the sources of Q3 and Q4. Thus, provided all transistors remain in the active mode, if Q3 and Q4 have identical gate voltages (irrespective of their absolute value), half of the (constant) tail current always flows through Q3 and half through Q4. Thus, Vin1 and Vin2 can vary, but if they have the same value, the voltage drops across the load resistors is constant and the output voltage does not change. Common mode gain is not zero because of the finite output resistance of Q2. If a differential voltage is placed between Vin1 and Vin2, then the fraction of the constant tail current that flows in Q3 and Q4 changes, causing a change in the voltage dropped across the load resistors.

(b) Calculate the following (you may ignore, i.e. assume infinite, the output resistance of the individual transistors when calculating quiescent conditions): [3+2+2]

i. The tail current and the drain current in transistors Q_3 and Q_4 in quiescent conditions.

ii. Calculate the value of R_D to set the output voltage at 12 V when $V_{in1} = V_{in2}$.

iii. Calculate the total quiescent power consumption.

Solution:

The tail current is given by: $I_D = K(V_{GS} - V_t)^2 = (5 - V_{GS})/3$
$\therefore V_{GS} = 2.18$ or 0.64
As V_{GS} must be above threshold, $V_{GS} = 2.18V$
$\therefore I_D = 0.94$ mA
And the drain current in Q_3 and Q_4 is 0.47 mA To set the quiescent output voltage to 12 V, we need to drop 8 V with the 0.47 mA, hence R_D =17.0 kΩ
The quiescent power consumption is $5 \times 0.94 + 20 \times 0.94 = 25.3$ mW

(c) Calculate the differential gain of the amplifier to the single ended output V_{out} when the differential voltage is defined as $V_{in1} - V_{in2}$. [3]

Solution: The single ended differential gain is given by:
$A_d = g_m r_o/2$
g_m of Q_3, (to which the output is connected), is $2\sqrt{kI_D} = 1.94$ mS
The output resistance of this transistor, $r_o = 300/0.47 = 638\ k\Omega$
The gain is therefore $-g_m.(r_o//R_D)/2$=-16.1

(d) It is suggested that Q_3 and Q_4 are replaced with BJTs, keeping all other components the same. What will the differential gain be? The early voltage of the proposed BJTs is the same as the MOSFETs. Comment on your answer. [5]

Solution: If the two transistors Q_3 and Q_4 are changes for BJTs, nothing changes in the quiescent condition of the circuit - because this is set by the current mirror (Q_1 and Q_2). The value of gm in the gain formula changes, and in this case to I_C/V_T where I_C is the same as the original drain current (0.47 mA) and V_T=25 mV.
$g_m = 0.47/25 = 18.8$ mS.
Thus the gain becomes $-18.8 \times 16.6/2 = -156$
The gain is significantly larger with the BJTs as the amplifying transistors - by a factor of the ratio of the transconductances. The transconductance of BJTs tends to be significantly larger than MOSFETS due to the exponential relationship between base-emitter voltage and current, compared to the square law relationship of the equivalent parameters for a MOSFET.

(e) It is proposed to reduce the output impedance of a source ($R_{out1} = R_S$), shown on the left of Figure 3e, by adding an emitter follower, shown on the right of Figure 3e. Draw a small signal model of the circuit on the right and derive an expression for its output impedance, R_{out2}. You may assume the BJT is in the forward active mode. [5]

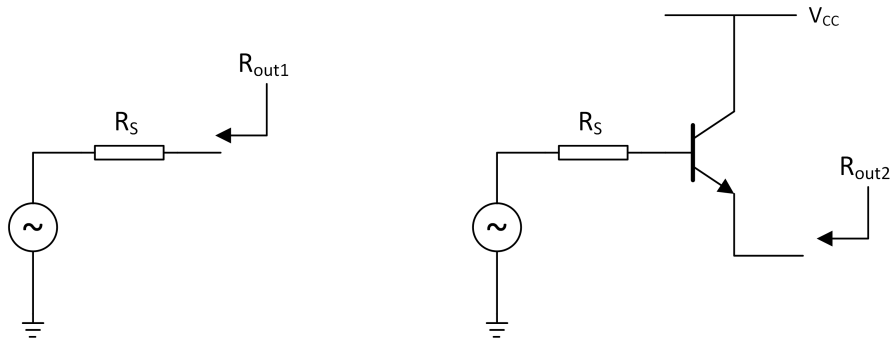
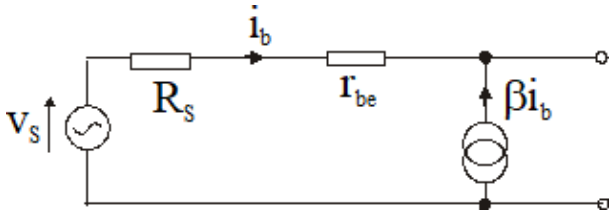


Figure 3e

Solution: The small signal model of the emitter follower is as follows:

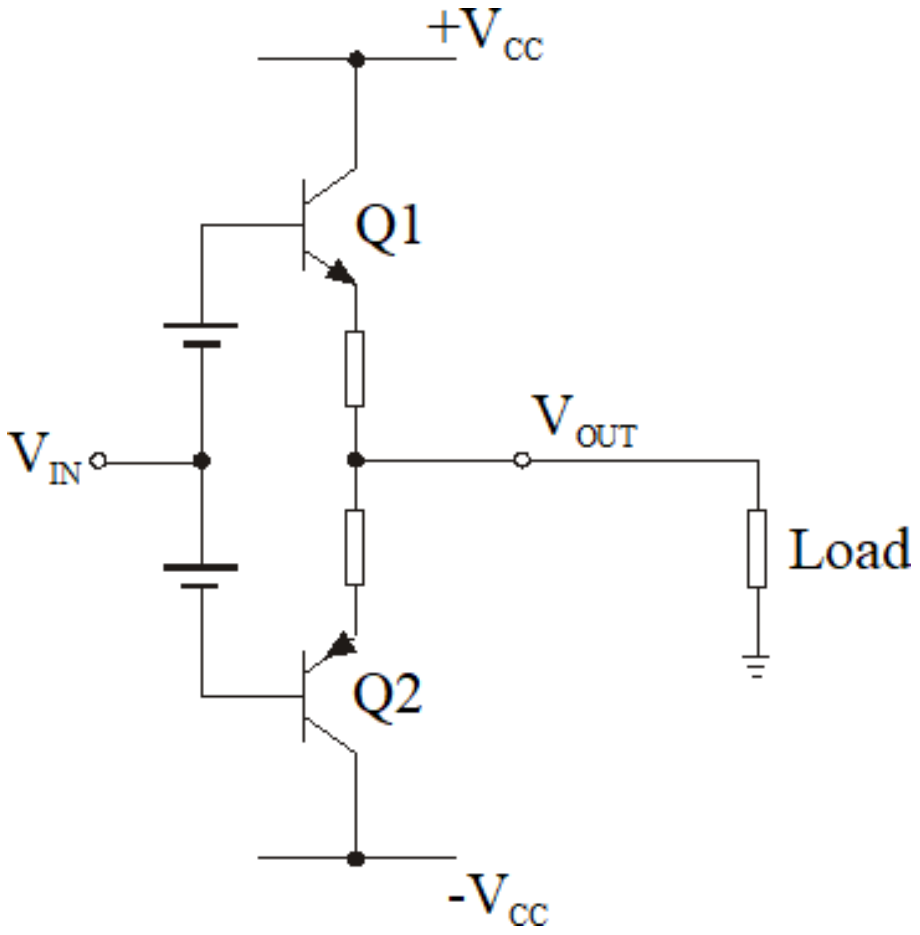


solution

Solution: The output current is: $(1+\beta)i_b$ When the input voltage is zero (to calculate output impedance),
$i_b = (-v_{out})/(R_s + r_{be})$
Hence $r_{out2} = (R_s + r_{be}/(1+\beta))$

(f) The emitter follower can only drive positive currents into the load. Suggest an alternative output stage that is able to drive both positive and negative currents into the load and explain its operation. [5]

Solution: The class B or class AB are suitable solutions that are able to drive both positive and negative currents. In both circuits the high side transistor conducts the positive half cycle and the lower transistor conducts the negative half cycle. The question does not ask specifically for a solution without cross-over distortion and so class B or AB are both legitimate answers. A class AB solution is shown below.



solution